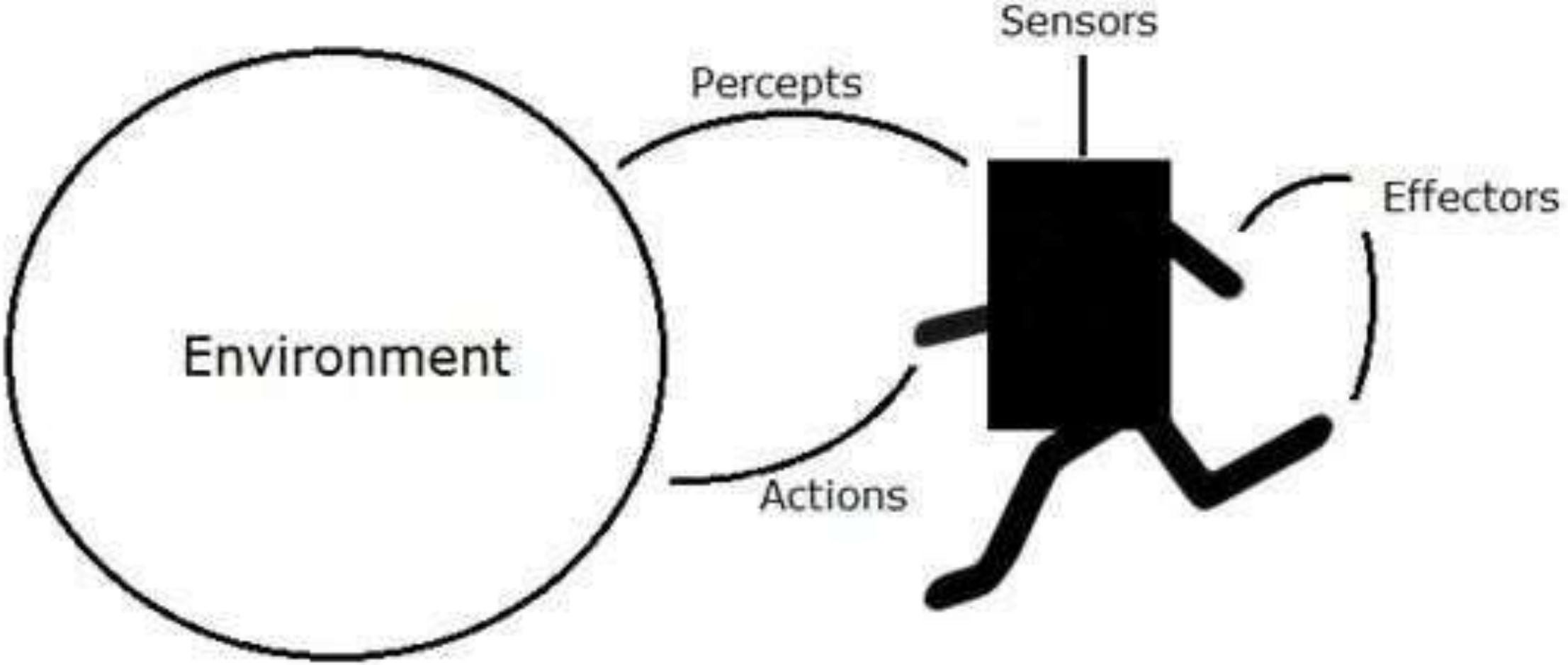
UNIT 1-Part 2

Agent

What are Agent and Environment?

- An **agent** is anything that can perceive its environment through **sensors** and acts upon that environment through **effectors.**
- A **human agent** has sensory organs such as eyes, ears, nose, tongue and skin parallel to the sensors, and other organs such as hands, legs, mouth, for effectors.
- A **robotic agent** replaces cameras and infrared range finders for the sensors, and various motors and actuators for effectors.
- A **software agent** has encoded bit strings as its programs and actions.


Sensors
Percepts
Effectors
Environment
Actions

Agent Terminology

- **Performance Measure of Agent** – It is the criteria, which determines how successful an agent is.
- **Percept** – It is agent's perceptual inputs at a given instance.
- **Percept Sequence** – It is the history of all that an agent has perceived till date.

Rationality

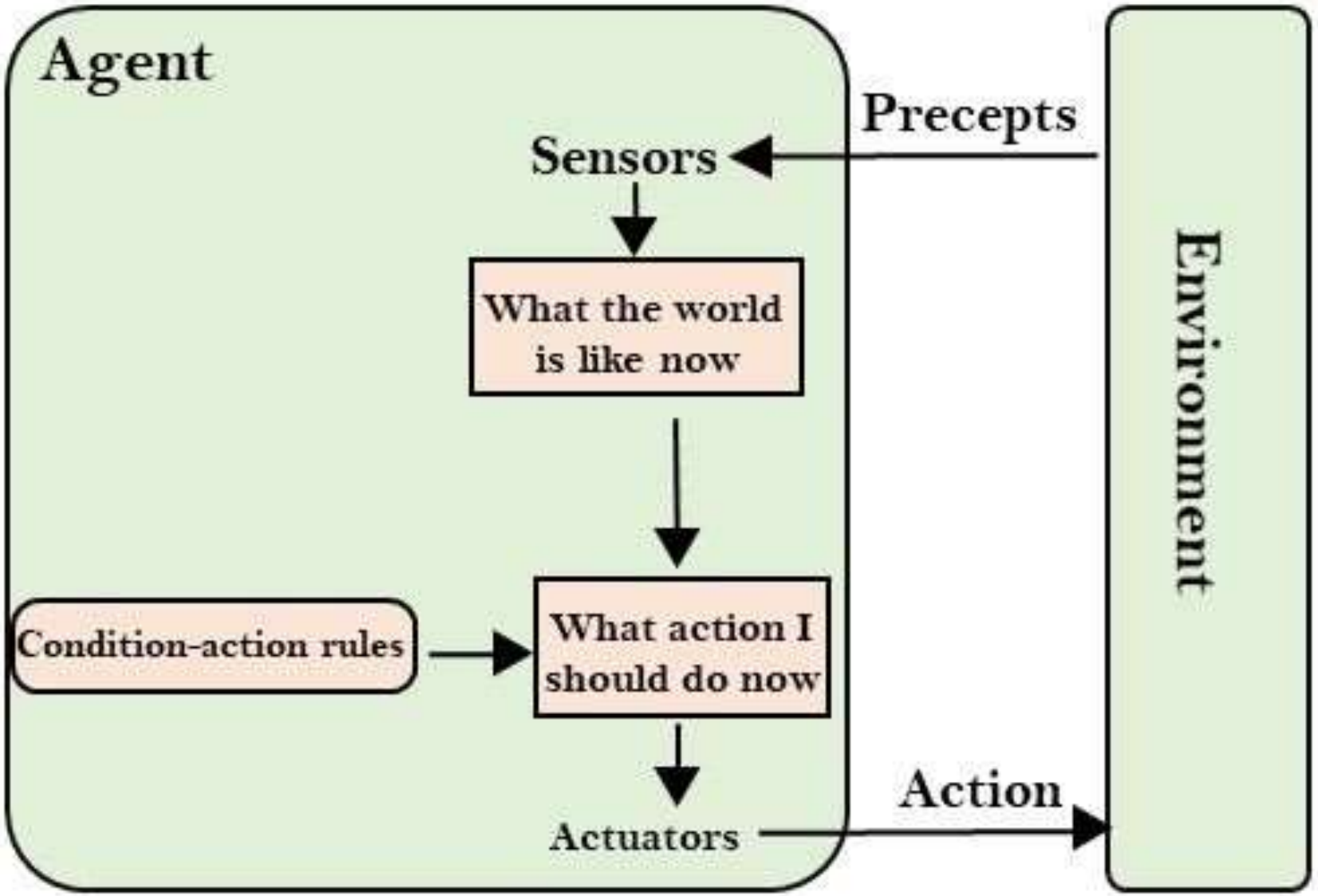
- Rationality is nothing but status of being reasonable, sensible, and having good sense of judgment.
- Rationality is concerned with expected actions and results depending upon what the agent has perceived. Performing actions with the aim of obtaining useful information is an important part of rationality.

Types of AI Agents

- Agents can be grouped into five classes based on their degree of perceived intelligence and capability. All these agents can improve their performance and generate better action over the time. These are given below:
- Simple Reflex Agent
- Model-based reflex agent
- Goal-based agents
- Utility-based agent
- Learning agent

Simple Reflex Agents

- They choose actions only based on the current percept.
- They are rational only if a correct decision is made only on the basis of current precept.
- Their environment is completely observable.
- The Simple reflex agent does not consider any part of percepts history during their decision and action process.
- The Simple reflex agent works on Condition-action rule, which means it maps the current state to action. Such as a Room Cleaner agent, it works only if there is dirt in the room.
- Problems for the simple reflex agent design approach:
  - They have very limited intelligence
  - They do not have knowledge of non-perceptual parts of the current state
  - Mostly too big to generate and to store.
  - Not adaptive to changes in the environment.

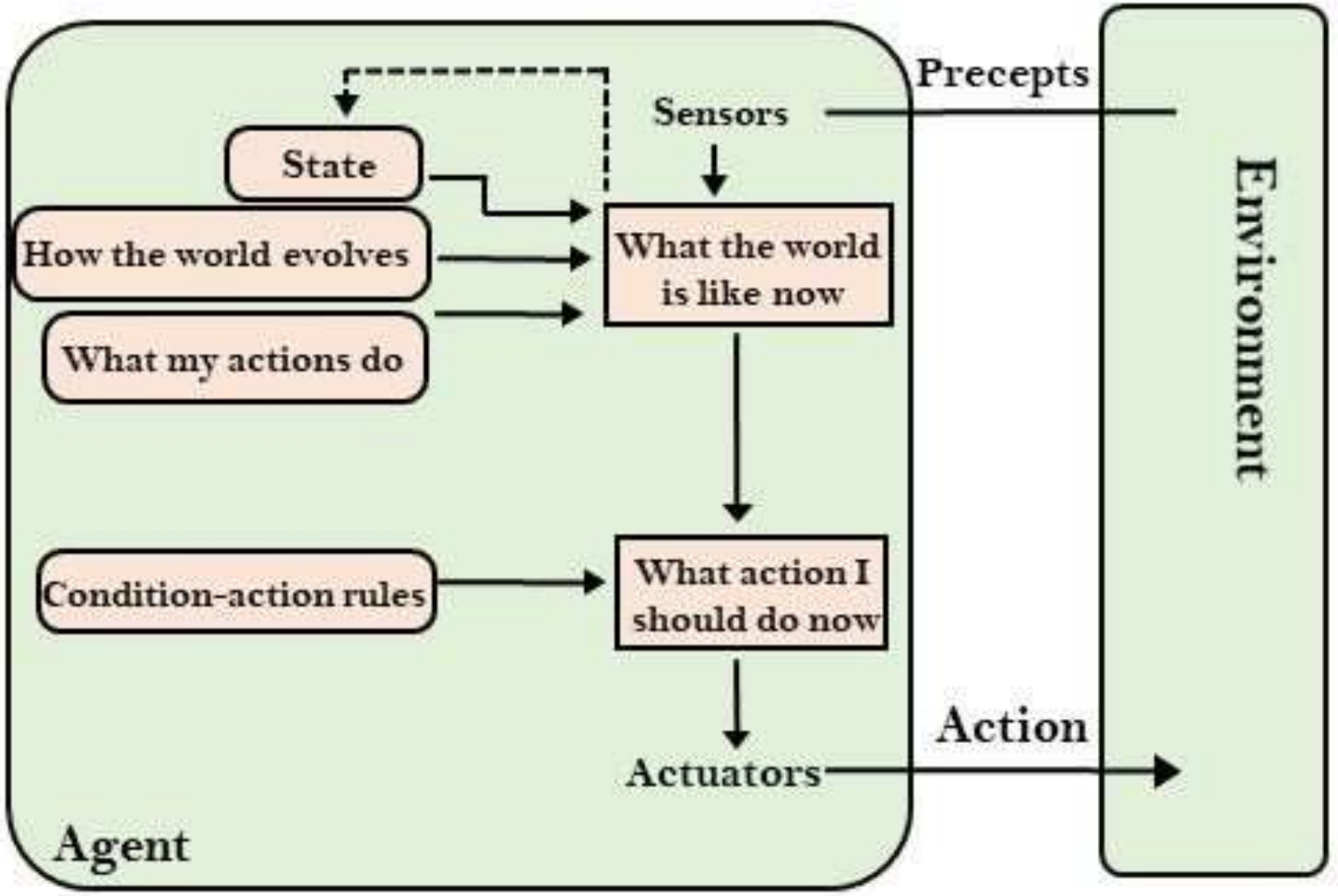

Agent
Precepts
Sensors
What the world is like now
Environment
Condition-action rules
What action I should do now
Action
Actuators

Simple Reflex Agents: Example

Automatic Door:
"If I (Agent) see someone nearby, open door."
Motion Sensor Light:
"If I (Agent) detect any movement, turn the light on."

2. Model-based reflex agent

- The Model-based agent can work in a partially observable environment, and track the situation.
- A model-based agent has two important factors:
  - **Model:** It is knowledge about "how things happen in the world," so it is called a Model-based agent.
  - **Internal State:** It is a representation of the current state based on percept history.
- These agents have the model, "which is knowledge of the world" and based on the model they perform actions.
- Updating the agent state requires information about:
  - How the world evolves
  - How the agent's action affects the world.

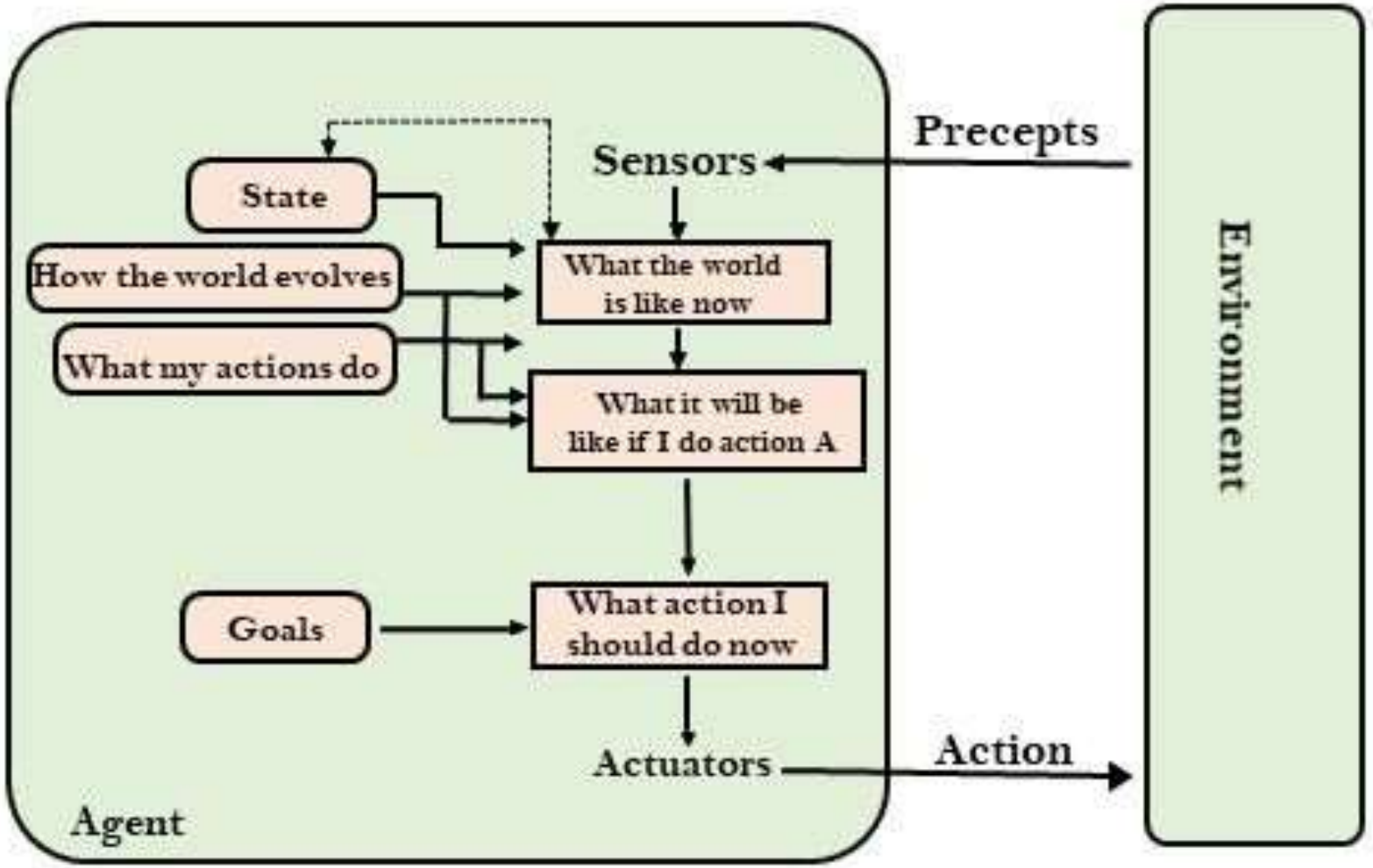

Precepts
Sensors
State
How the world evolves
What my actions do
What the world is like now
Environment
Condition-action rules
What action I should do now
Action
Actuators
Agent

Model-Based Reflex Agents: Example

Example: Remembering a User's Name
User:
"Call me Smrity."
Simple Reflex Agent:
"I don’t know. You didn’t tell me in this message."
Model-Based Reflex Agent:
"Your name is Smrity!"
Explanation:
A simple reflex agent only uses the current input and does not remember past information.
A model-based reflex agent maintains an internal memory (model) of previous interactions and uses that information to make decisions.

3. Goal-based agents

- The knowledge of the current state environment is not always sufficient to decide for an agent to what to do.
- The agent needs to know its goal which describes desirable situations.
- Goal-based agents expand the capabilities of the model-based agent by having the "goal" information.
- They choose an action, so that they can achieve the goal.
- These agents may have to consider a long sequence of possible actions before deciding whether the goal is achieved or not. Such considerations of different scenario are called searching and planning, which makes an agent proactive.


Precepts
Sensors
State
How the world evolves
What my actions do
What the world is like now
What it will be like if I do action A
Goals
What action I should do now
Actuators
Action
Agent
Environment

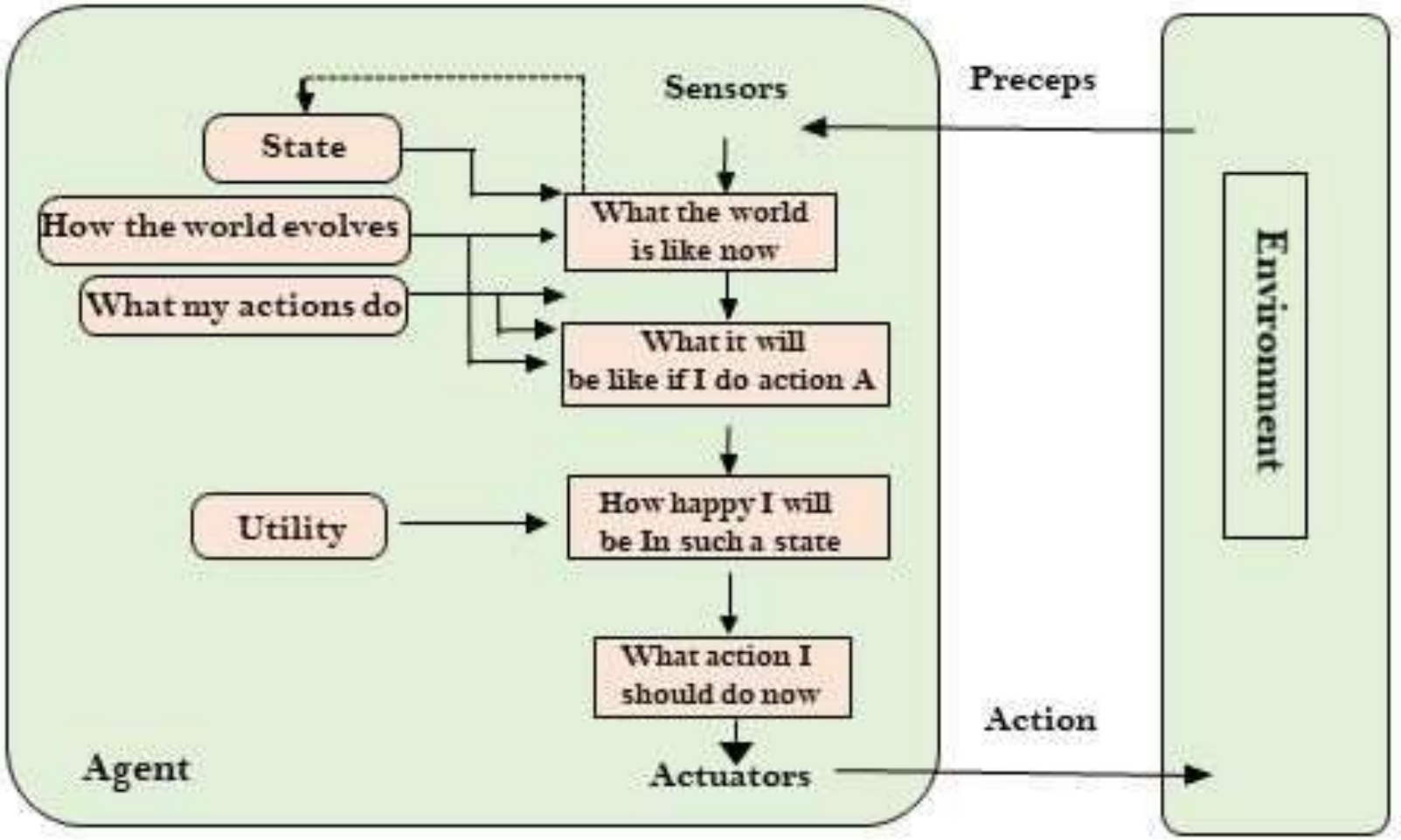
Goal-Based Agents:Example

Example: Navigation App (Google Maps)

- Your goal is to reach the destination.
- It checks roads, traffic, and suggests the best route.
- If one road is closed, it replans a new path.
- Explanation:
  A goal-based agent takes actions based on a specific goal. It considers different possible actions and chooses the best path to achieve the desired outcome.

4. Utility-based agents

- These agents are similar to the goal-based agent but provide an extra component of utility measurement which makes them different by providing a measure of success at a given state.
- Utility-based agent act based not only goals but also the best way to achieve the goal.
- The Utility-based agent is useful when there are multiple possible alternatives, and an agent has to choose in order to perform the best action.
- The utility function maps each state to a real number to check how efficiently each action achieves the goals.

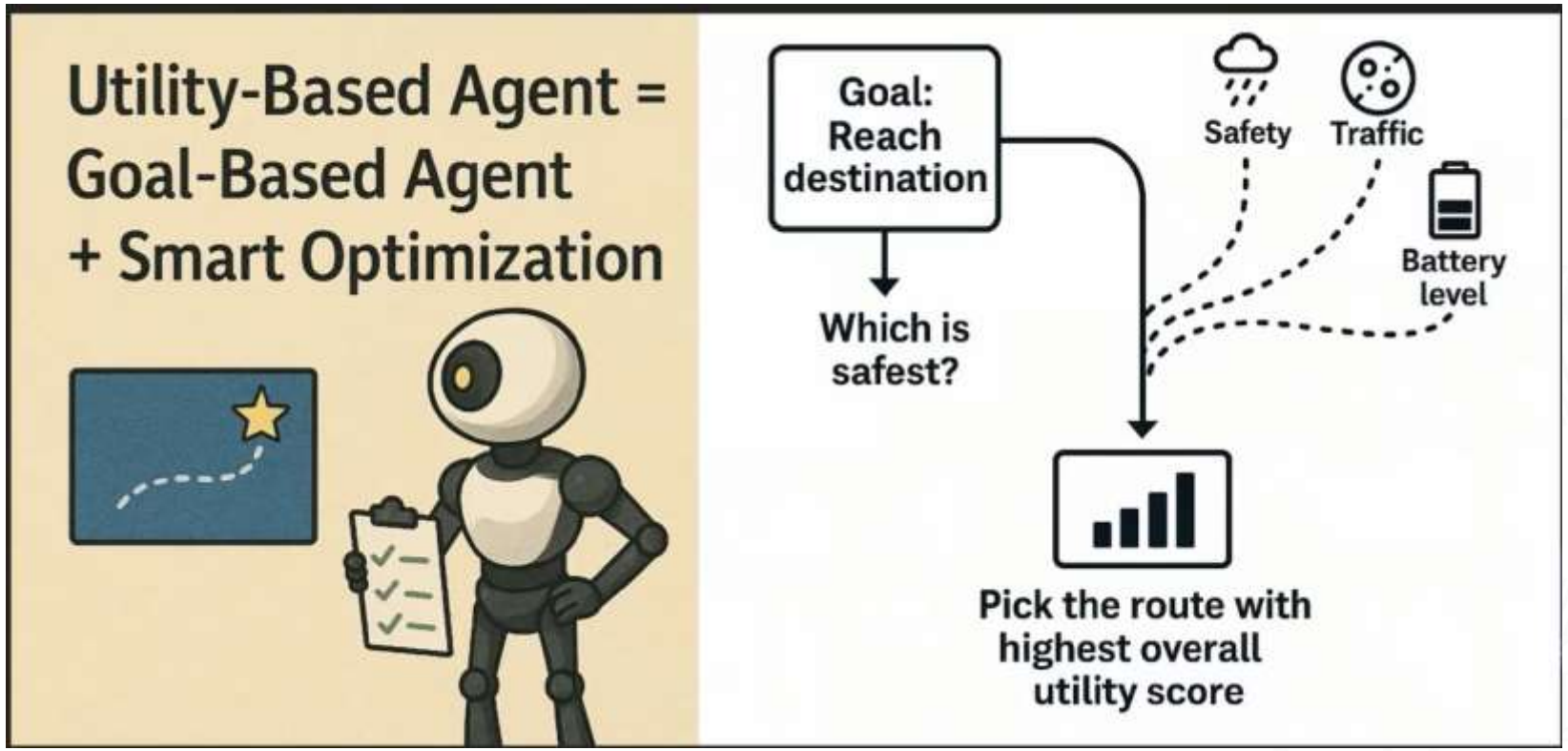

Sensors
Preceps
State
How the world evolves
What my actions do
What the world is like now
What it will be like if I do action A
Utility
How happy I will be In such a state
What action I should do now
Action
Agent
Actuators
Environment

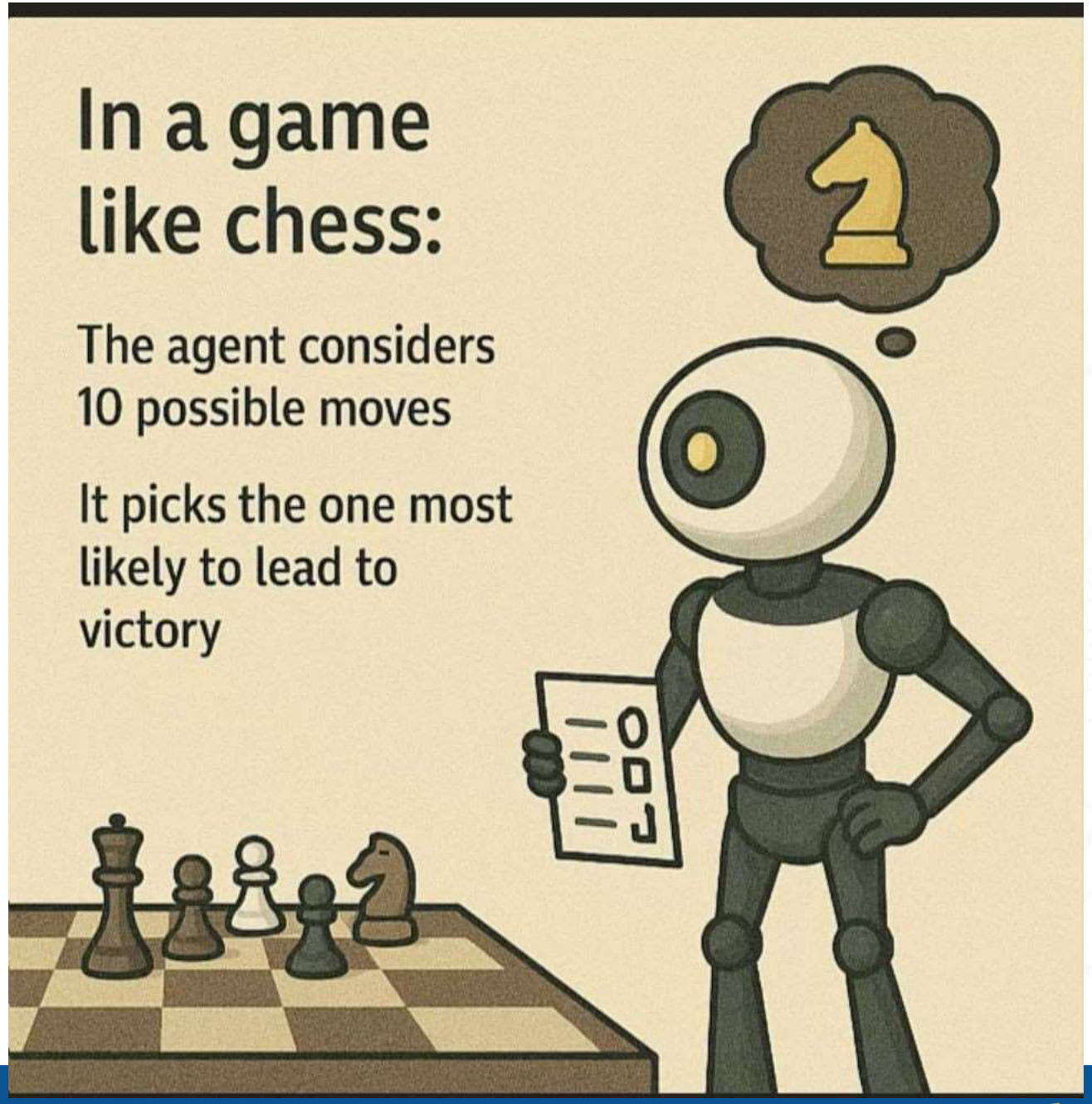
Utility-Based Agents:Example

Utility-Based Agent =
Goal-Based Agent + Smart Optimization
Example: Flight Booking Agent
Goal:
Book a flight under ₹5000.
The agent compares different options:

- ✈️ **Flight 1:** ₹4500
  Utility Score: 6
- ✈️ **Flight 2:** ₹5200
  Utility Score: 9
- ✈️ **Flight 3:** ₹4800
  Utility Score: 7

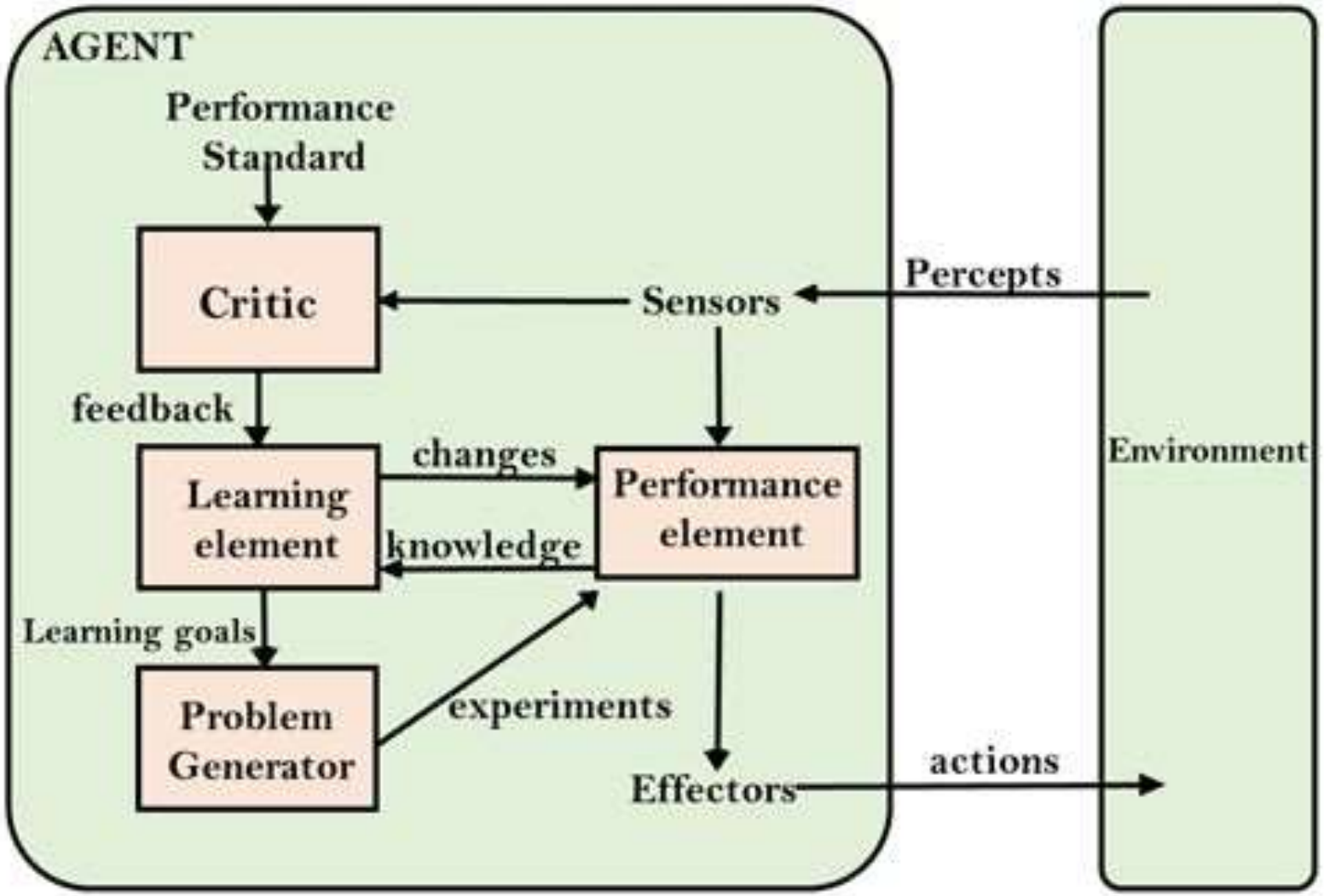
Explanation:
A utility-based agent not only tries to achieve a goal but also evaluates different choices and selects the option that provides the highest benefit or utility. It considers factors like cost, quality, time, and user preferences to make the best decision.


Utility-Based Agent =
Goal-Based Agent
+ Smart Optimization
Goal:
Reach
destination
Which is
safest?
Safety
Traffic
Battery
level
Pick the route with
highest overall
utility score


In a game like chess:
The agent considers 10 possible moves
It picks the one most likely to lead to victory
LOL

5. Learning Agents

- A learning agent in AI is the type of agent which can learn from its past experiences, or it has learning capabilities.
- It starts to act with basic knowledge and then able to act and adapt automatically through learning.
- A learning agent has mainly four conceptual components, which are:
  - **Learning element:** It is responsible for making improvements by learning from environment
  - **Critic:** Learning element takes feedback from critic which describes that how well the agent is doing with respect to a fixed performance standard.
  - **Performance element:** It is responsible for selecting external action
  - **Problem generator:** This component is responsible for suggesting actions that will lead to new and informative experiences.
- Hence, learning agents are able to learn, analyse performance, and look for new ways to improve the performance.


AGENT
Performance Standard
Critic
Sensors
Percepts
feedback
changes
Learning element
Performance element
knowledge
Environment
Learning goals
Problem Generator
experiments
Effectors
actions

Learning Agents: Example

A Learning Agent
is an agent that **gets better over time.**
It learns from its own experiences –
just like we do!
Learning Process:
Experience → Feedback → Update Knowledge
Explanation:
A learning agent improves its performance by learning from past experiences and feedback. It updates its knowledge and uses what it has learned to make better decisions in the future.

Good Behavior:

In artificial intelligence, **good behavior** is defined through the concept of **rationality**,

Good Behavior: The Concept of Rationality

- Rationality is the standard used to measure the intelligence of an agent. A system is considered rational if it does the **"right thing,"** given what it knows through its percepts and internal knowledge.

- **Performance Measures:** Rational behavior is evaluated by an **objective performance measure**, which is a criterion for success typically imposed by the agent's designer. It is generally better to design these measures based on the desired state of the environment (e.g., a clean floor) rather than how the agent should behave (e.g., cleaning up dirt, then dumping it to clean it again).
- **The Rational Agent:** Formally, a rational agent is one that, for each possible percept sequence, selects an action **expected to maximize its performance measure**, based on the evidence from the percept sequence and its built-in knowledge.

- **Rationality vs. Perfection:** Rationality is not the same as perfection. Perfection maximizes *actual* performance (which requires impossible omniscience), whereas rationality maximizes ***expected* performance** based on available information.
- **Information Gathering and Exploration:** A rational agent should perform actions to modify future percepts, such as looking both ways before crossing a street. This is known as **information gathering** or **exploration** (especially in unknown environments).

Environments

world or surroundings where an AI agent operates, perceives its environment, and takes actions.

If the AI is a self-driving car, the environment is the busy city street. If the AI is a chess-playing bot, the environment is the 64-square chessboard.

Task Environment in Artificial Intelligence

A task environment is the specific situation or problem in which an intelligent agent operates. It defines everything that the agent needs to know and interact with to perform its task successfully.
In simple words, a task environment describes the world around an AI agent and the goal it must achieve.
A task environment includes:

- What the agent needs to achieve
- What information it receives
- What actions it can perform
- What conditions exist in the environment
- The task environment is usually described using the PEAS framework:

The Nature of Environments

Environments are classified by their **nature** using several key properties. Understanding these properties is crucial because they dictate exactly what kind of AI algorithm you need to build.

Environment Property	What it Means	Real-World Example
Fully Observable vs. Partially Observable	Fully Observable: The AI can access all information about the current state of the environment. Partially Observable: Some information is hidden, missing, or unavailable due to limited sensors or uncertainty.	Fully Observable: Chess — the AI can see the complete chessboard and all pieces. Partially Observable: Poker — the AI cannot see the opponent's hidden cards.
Deterministic vs. Stochastic	Deterministic: The next state of the environment is completely determined by the current state and the agent's action. Stochastic: The outcome is uncertain and may involve randomness or unexpected events.	Deterministic: Calculator — the same input always produces the same output. Stochastic:Driving a car — weather, traffic, and other drivers create uncertainty.

Environment Property	What it Means	Real-World Example
Static vs. Dynamic	Static: The environment does not change while the AI is deciding its action. Dynamic: The environment changes continuously while the AI is making decisions.	Static: Crossword puzzle — the puzzle remains unchanged while solving. Dynamic: Drone navigation or soccer — the environment changes continuously.
Discrete vs. Continuous	Discrete: The environment has a limited number of possible states, actions, or steps. Continuous: States and actions can change smoothly over time with many possible values.	Discrete: Board games like chess — fixed moves and positions. Continuous: Car driving — speed, location, and steering angle change continuously.

Environment Property	What it Means	Real-World Example
Single-Agent vs. Multi-Agent	Single-Agent: Only one intelligent agent operates in the environment. Multi-Agent: Multiple agents interact, either competing or cooperating with each other.	Single-Agent: Crossword puzzle solver — works independently. Multi-Agent:Multiplayer games or stock trading systems where multiple agents interact.
Known vs. Unknown	Known: The agent already knows the rules and how the environment works. Unknown: The agent must explore and learn the rules of the environment through experience.	Known: A video game where the rules and physics are already defined. Unknown: A robot exploring an unfamiliar deep-sea environment.

Environment Property	What it Means	Real-World Example
Episodic vs. Sequential	Episodic: Each action is independent, and one decision does not affect future decisions. Sequential: Current actions affect future states and decisions.	Episodic: A factory robot checking products on a conveyor belt — each product inspection is independent. Sequential: Chess or video games — current moves affect future outcomes.

PEAS Representation

- PEAS is a type of model on which an AI agent works upon. When we define an AI agent or rational agent, then we can group its properties under PEAS representation model. It is made up of four words:
- **P:** Performance measure
- **E:** Environment
- **A:** Actuators
- **S:** Sensors
- Here performance measure is the objective for the success of an agent's behaviour.

PEAS for self-driving cars:

- Let's suppose a self-driving car then PEAS representation will be:

- **Performance:** Safety, time, legal drive, comfort
- **Environment:** Roads, other vehicles, road signs, pedestrian
- **Actuators:** Steering, accelerator, brake, signal, horn
- **Sensors:** Camera, GPS, speedometer, odometer, accelerometer, sonar.

Example of Agents with their PEAS representation

Agent	Performance measure	Environment	Actuators	Sensors
1. Medical Diagnose	•Healthy patient •Minimized cost	•Patient •Hospital •Staff	•Tests •Treatments	Keyboard (Entry of symptoms)
2. Vacuum Cleaner	•Cleanness •Efficiency •Battery life •Security	•Room •Table •Wood floor •Carpet •Various obstacles	•Wheels •Brushes •Vacuum Extractor	•Camera •Dirt detection sensor •Cliff sensor •Bump Sensor •Infrared Wall Sensor
3. Part -picking Robot	•Percentage of parts in correct bins.	•Conveyor belt with parts, •Bins	•Jointed Arms •Hand	•Camera •Joint angle sensors.